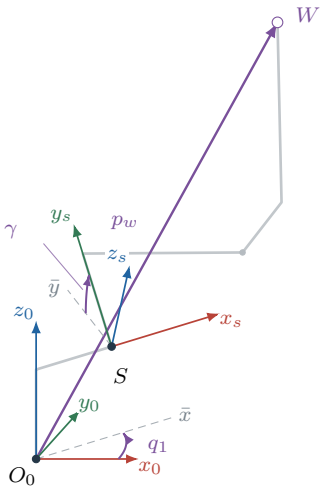
From the physical arm to the position equations

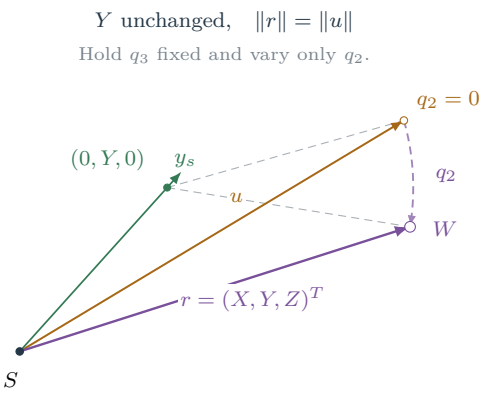
The same configuration is used throughout: $q_1 = 40°$, $q_2 = 25°$, $q_3 = 75°$ and $\gamma = 30°$. Each panel isolates one geometric idea used in the derivation.

(a) Locating the shoulder frame



The shoulder frame has origin S. Its orientation includes q_1 and the fixed roll γ, which tilts $\bar{y} = R_z(q_1)e_y$ into y_s. It is independent of q_2.

(b) Why q_2 can be eliminated

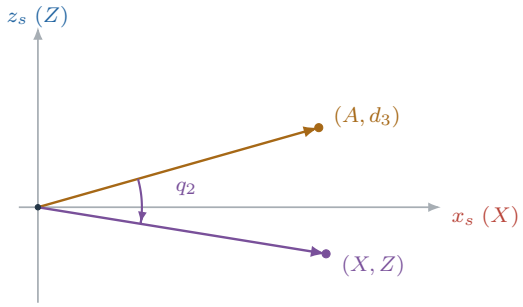


In the shoulder frame, u is the wrist vector at $q_2 = 0$ and $r = R_y(q_2)u$. The rotation preserves the axial coordinate Y and the squared distance $N = \|r\|^2$.

$$p_w \xrightarrow{\text{Undo the base rotation}} R_z(-q_1)p_w \xrightarrow{\text{Move the origin to } S} R_z(-q_1)p_w - s \xrightarrow{\text{Undo the fixed roll}} r = (X, Y, Z)^T$$

$$s = (a_1, 0, d_1)^T, \qquad r = R_x(-\gamma)\big(R_z(-q_1)p_w - s\big).$$

(c) Recovering q_2 in the $x_s z_s$ plane



Positive q_2 is clockwise in this projection.

With $A = a_2 + d_4 c_3$, the reference vector is $u = (A, Y, d_3)^T$. Panel (c) drops the common y_s component. The planar rotation gives

$$X = A\cos q_2 + d_3 \sin q_2,$$
$$Z = -A\sin q_2 + d_3 \cos q_2.$$

Combining these equations isolates a scaled cosine and sine:

$$AX + d_3 Z = (A^2 + d_3^2)\cos q_2,$$
$$d_3 X - AZ = (A^2 + d_3^2)\sin q_2.$$

Their common factor $A^2 + d_3^2$ is positive, so applying atan2 to these scaled sine and cosine values gives

$$q_2 = \operatorname{atan2}(d_3 X - AZ,\ AX + d_3 Z).$$

Geometry and notation. Point locations and frame orientations follow the CuRo6R URDF. The faint arm outline provides context; panels (b) and (c) show the vectors needed for the derivation. Gold denotes the reference vector at $q_2 = 0$ and purple denotes the vector after the shoulder rotation.